النظم الذكية
1:00-2:00
6/4/2025
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Faculty of Computers & Information, Assiut University
Course code: (CS461
4th Level
Midterm Exam
Duration: One hour
This exam for the following program(s) : (Add the program's name(s))

Points: -/30

1. الإسم الرباعي (بالعربي فقط) *

2. رقم الجلوس *

٤د

٤هـ

6. رقم الكمبيوتر *

7. الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) *

8. Which search strategy is also called as blind search? *

Uninformed search

Informed search

Simple reflex search

All of the above

9. Which of the following best describes "machine learning"? *

A method for programming computers to perform tasks

A technique that allows systems to learn from data and improve over time

A type of artificial intelligence that does not require data

A process that only uses human input for decision-making

10. In the context of Iterative Deepening Depth First Search, what does the term "depth" refer to? *

The number of nodes in the search tree.

The maximum number of actions taken to reach a goal state.

The maximum level of the search tree explored in a single iteration.

The total number of iterations performed.

11. Simulated annealing at low temperatures primarily focuses on *

A) Simulation

B) Intensification

C) Diversification

A and B

12. What happens to the temperature in Simulated Annealing as the algorithm progresses? *

It remains constant

It increases exponentially

It gradually decreases

None of the above

13. Which search is implemented with an empty first-in-first-out queue? *

Depth-first search

Bidirectional search

Breadth-first search

None of the above

14. Which of the following components is NOT typically part of an Intelligent system? *

Knowledge base

Inference engine

User interface

Operating system

15. How many successors are generated in backtracking search? *

1

2

3

4

16. Which search implements stack operation for searching the states? *

Depth-limited search

Depth-first search

Breadth-first search

None of the above

17. The search that uses an evaluation function to determine the best node to visit next is known as *

Depth-First Search

Breadth-First Search

A* Search

None of the above

18. What is the primary goal of Tabu Search? *

To improve local solutions

To reduce execution time

To increase the number of possible solutions

To avoid revisiting previously explored solutions

19. _______ is the ability to derive specific conclusions from general principles or rules *

Deductive Reasoning

Inductive Reasoning

Abductive Reasoning

None of the above

20. A search algorithm takes _________ as an input and returns ________ as an output. *

Input, output

Problem, solution

Solution, problem

Parameters, sequence of actions

21. In Backward Bidirectional Search, the search starts from *

The initial state and moves towards the goal state.

The goal state and moves towards the initial state.

A random state in the search space.

Both the initial and goal states simultaneously.

22. In Simulated Annealing, what role does the temperature parameter play? *

It determines the number of iterations.

It defines the search space.

It controls the probability of accepting worse solutions.

It sets the stopping criteria.

23. What is the primary characteristic of Uniform Cost Search? *

It explores nodes in a depth-first manner.

It explores all nodes at the current depth before moving deeper.

It uses a heuristic to guide the search.

It expands the node with the lowest path cost.

24. The type of search that visits all nodes at one level before moving on to the next level is called: *

Breadth-First Search (BFS)

Depth-First Search (DFS)

Best-First Search

Uniform Cost Search

25. A problem in a search space Is defined by *

Initial state

Goal test

Intermediate states

All of the above

26. In the context of intelligent systems, what does "adaptivity" refer to? *

The ability to perform tasks without any data

The ability to change behavior based on new information or environments

The ability to store large amounts of data

The ability to execute predefined commands

27. The Set of actions for a problem in a state space is formulated by a ___________. *

Intermediate states

Initial state

Successor function, which takes current action and returns next immediate state

None of the above

28. What is a balanced tree? *

All nodes have the same number of children

The difference in height between any two branches does not exceed a certain value

It contains only one root node

It can have an unlimited number of children

29. Which of the following statements is true about Simulated Annealing? *

It guarantees finding the global optimum.

It can escape local minima by accepting worse solutions.

It is a type of greedy algorithm that always chooses the best immediate solution.

None of the above

30. When does the Hill Climbing algorithm terminate? *

When a global maximum is found

When no neighboring state has a higher value

After a fixed number of iterations

All of the above

31. Which of the following is a limitation of the Hill Climbing algorithm? *

It can find the global optimum.

It may get stuck in local maxima.

It is computationally not efficient.

It requires a heuristic function.

32. What is the meaning of "evaluation" of the node in the tree? *

Determining the number of nodes in the tree

Improving the performance of the algorithm

Reducing the number of possible solutions

Estimating the value of a node based on the current state

33. What is the depth limit in the first iteration of Iterative Deepening Depth First Search? *

0

1

2

It depends on the problem.

34. Which of the following is an example of an Intelligent system? *

A spreadsheet application

A web browser

A word processing program

A medical diagnosis system

35. Which of the following is true about the successor function? *

It can only return one state for each action.

It can return multiple states for a single action.

It is not necessary for defining a search problem.

None of the above

36. What is the main characteristic of Hill Climbing? *

It always chooses the neighbor with the highest value.

It randomly selects neighbors.

It terminates after a set number of moves.

None of the above

37. What is the effect of a high initial temperature in Simulated Annealing? *

It allows for more exploration of the solution space.

It leads to be toward the greedy search.

It reduces the chances of accepting worse solutions.

It terminates the algorithm faster.